Synthetic Paper B: Graph-Guided Retrieval for Multi-Hop Questions

Fixture ID SYN-B - Version 1.0 - Intended for project-level retrieval and paper comparison

TEST DATA DISCLOSURE: This paper is fictional. All study descriptions, sample sizes, measurements, and results below are invented solely to test MAIROS workflows. Do not cite as published research.

Abstract

This fictional study describes graph-guided retrieval for questions requiring connections between multiple concepts. It contrasts dense retrieval alone with dense retrieval followed by a small entity-link traversal. Its invented results and limitations give the MAIROS graph, task, comparison, and research-gap workflows structured content to find.

Research question and hypothesis

RQ-B: Does adding graph-neighbor passages to dense retrieval improve Recall@5 for multi-hop questions, and what latency trade-off does it introduce? H-B: graph-guided retrieval increases Recall@5 while increasing median latency. The claim is intentionally narrow and fictional.

Methods

The fixture uses 80 synthetic multi-hop questions and 40 synthetic paper summaries. Dense-only ranks passages by embedding similarity. Graph-guided retrieval first takes dense top-3 results, follows one entity/concept edge, adds up to two neighboring passages, then reranks the candidate set. Both methods use the same question set, source summaries, and top-five evaluation cutoff.

Dataset and measures

Invented measures are Recall@5 (whether a gold evidence passage appears among the first five results) and median retrieval latency in seconds. The described graph has manually assigned synthetic concept and paper nodes. It does not represent a real knowledge graph or real indexed literature.

Synthetic Paper B: Graph-Guided Retrieval for Multi-Hop Questions — results and interpretation

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Measure (invented)	Dense-only	Dense + graph neighbors
Recall@5	0.62	0.78
Median latency	0.42 s	0.68 s
Gold evidence found	50 / 80 questions	62 / 80 questions

Illustrative interpretation

In this fictional fixture, graph expansion finds more gold evidence passages within five results and takes longer. The numbers are internally aligned (62/80 = 0.775, rounded to 0.78). They exist to make the workflow compare meaningful fields and to help inspect concept-to-paper links.

Limitations and open questions

The graph is small and manually constructed; entity-link mistakes, missing edges, and graph incompleteness are not measured. Only one traversal depth and one top-five cutoff are described. Open questions include whether the recall gain remains when edges are noisy, whether a two-hop traversal adds enough coverage to justify its latency, and whether gains hold on independently authored real papers.

Comparison hooks

SYN-B focuses on cross-concept retrieval and graph expansion; SYN-A focuses on claim support and citation entailment; SYN-C focuses on chunk boundaries. Treat the project knowledge graph as a map of indexed fixture content, not proof that a topic is absent from the broader research field.